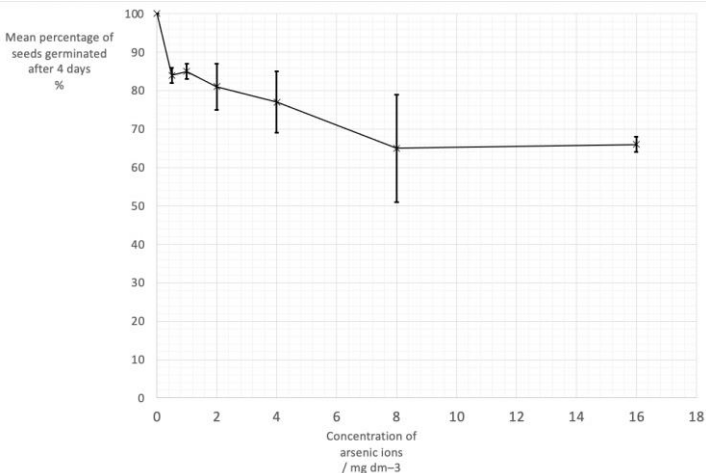


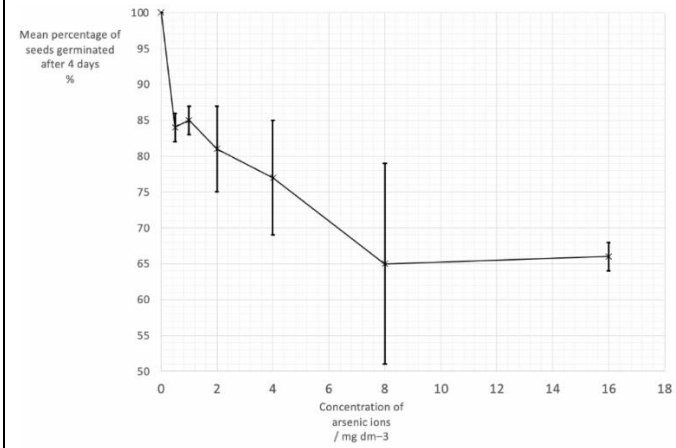
Question Number	Answer	Additional Guidance	Mark
3(a)(i)	• suitable risk identified (1) • suitable control identified (1)	Any reasonable risk identified eg toxic effects, allergy, irritant ignore infection/labcoat/PPE Any corresponding control identified eg use of gloves/mask	Exp (2)

Question Number	Answer	Additional Guidance	Mark
3(a)(ii)	• there is no (significant) correlation between the (mean) percentage of germination (after 4 days) and the concentration of arsenic ions		Exp (1)

Question Number	Answer	Additional Guidance	Mark																								
3(a)(iii)	An answer that includes three of the following: - axes labelled including units (1) - data plotted correctly on a linear scale in scatter graph format (1) - error bars correctly plotted (1) <table><tr><th>concentration /mg dm⁻³</th><th>mean %</th><th>error bar</th></tr><tr><td>0.0</td><td>100</td><td></td></tr><tr><td>0.5</td><td>84 ± 2</td><td>82 - 86</td></tr><tr><td>1.0</td><td>85± 2</td><td>83 - 87</td></tr><tr><td>2.0</td><td>81± 6</td><td>75 - 87</td></tr><tr><td>4.0</td><td>77 ± 8</td><td>69 - 85</td></tr><tr><td>8.0</td><td>65 ±14</td><td>51 - 79</td></tr><tr><td>16.0</td><td>66 ± 2</td><td>64 - 68</td></tr></table>	concentration /mg dm ⁻³	mean %	error bar	0.0	100		0.5	84 ± 2	82 - 86	1.0	85± 2	83 - 87	2.0	81± 6	75 - 87	4.0	77 ± 8	69 - 85	8.0	65 ±14	51 - 79	16.0	66 ± 2	64 - 68	Accept y axis mean % germination and x axis concentration of arsenic ions/ mg dm⁻³ Accept broken scale with symbol or scale not starting at zero Ignore line if drawn Example graph 	
concentration /mg dm ⁻³	mean %	error bar																									
0.0	100																										
0.5	84 ± 2	82 - 86																									
1.0	85± 2	83 - 87																									
2.0	81± 6	75 - 87																									
4.0	77 ± 8	69 - 85																									
8.0	65 ±14	51 - 79																									
16.0	66 ± 2	64 - 68																									

Exp
(3)

Example graph y axis not at zero



Question Number	Answer	Additional Guidance	Mark
3(b)(i)	- calculation of intermediate step (1) - calculation of r_s (1)	Accept $\sum d^2 = 24$ and $n=7$ OR $\frac{6 \sum d^2}{n(n^2 - 1)} = \frac{24}{336}$ 13/14 $r_s = 0.929 / 0.93 / 0.9/0.9286$ must be correct rounding correct answer with no working gains full marks	Exp (2)

Question Number	Answer	Additional Guidance	Mark
3(b)(ii)	An answer that includes three of the following: - calculated value (0.93) is greater than the critical value, 0.786 therefore reject the null hypothesis (1) - there is a (significant) correlation between the percentage germination and the concentration of arsenic ions (1) - comment on variability of data (1)	Accept $0.93 > 0.786$, so reject the null hypothesis Accept critical value identified in table only Accept a correct statement using the critical value from $p = 0.01$ Accept converse wording for MP1 And 2 if they claim the calculated value is less than.... Eg error/range bars overlap	Exp (3)

Question Number	Answer	Additional Guidance	Mark
3(c) (i)	An answer that includes two of the following: • differences in {alleles/DNA} (may alter the response to arsenic ions) (1) • some varieties may be unable to <u>absorb</u> arsenic ions (during germination/in 4 days) (1) • reference to mutation/natural selection (1)	Ignore genes Accept comment on membrane permeability Accept details of one form of mutation	Exp (2)

Question Number	Answer	Additional Guidance	Mark
3(c)(ii)	An answer that includes three of the following: • germinate seeds with no arsenic (1) • (grown in) different concentrations of arsenic ions (1) • control of a stated condition (1) • measure {height/mass} of plants after {stated/same length of time} (1)	Ignore soil Eg temperature, volume of water, light , pH accept other suitable parameter eg 7-100 days	Exp (3)

(Total for question 3 = 16 marks)